

Week 1-3 Order statistics (6.2-6.3)

Def for $X_1 \dots X_n$ data sample
order statistics are
 $Y_1 = \text{smallest of } X_1, \dots, X_n$
 $Y_2 = \text{2nd smallest of } X_1, \dots, X_n$
 $\vdots$
 $Y_n = \text{largest of } X_1, \dots, X_n$

Ex Test scores (out of 100)

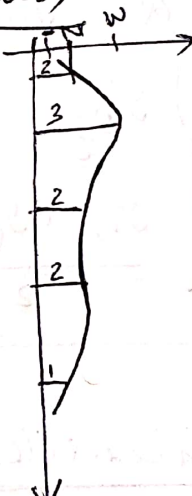
93	77	67	72	52	83	66	84	59	63
X_1	X_2	...							X_n

$n=10$

① Order data samples from smallest to largest

One way to do it: stem-and-leaf display

Stems	Leaves	(frequencies)
5	2 9	2
6	7 6 3	3
7	7 2	2
8	3 4	2
9	3	1



pp 238-240
versions of
stem-and-leaf

52	59	63	66	67	72	77	83	84	93
Y_1	Y_2	...							Y_n

order statistics

② Find median = 50th percentile
= bigger than $\approx 50\%$ of all data

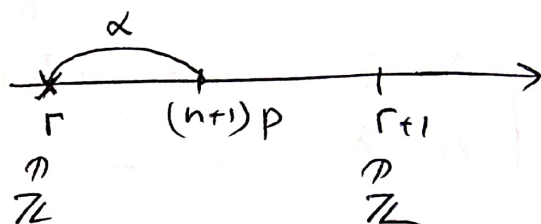
Is it 67, or 72, or smth in between?

If 11 scores, we would take $y_6 = y_{(n+1) \cdot \frac{1}{2}}$
 This motivates

Def To find $(100p)$ -percentile, take number of samples

$$\textcircled{*} \quad \tilde{\pi}_p = \begin{cases} y_{(n+1) \cdot p} & \text{order statistics, if } (n+1)p \in \mathbb{Z} \\ y_r + \alpha (y_{r+1} - y_r), & \text{if } (n+1)p \notin \mathbb{Z} \end{cases}$$

$$r = \lfloor (n+1)p \rfloor \quad \alpha = (n+1)p - r$$



medians: $\tilde{m} = \tilde{\pi}_{0.5} = y_5 + \frac{1}{2} y_6 = 67 + \frac{1}{2} (72 - 67) = \textcircled{69.5}$

For $\tilde{m}$, definition $\textcircled{*}$ takes the form

$$\tilde{\pi}_{0.5} = \tilde{m} = \begin{cases} y_{\frac{(n+1)}{2}} & \text{if } n \text{ is odd} \\ \frac{y_{n/2} + y_{(n/2+1)}}{2} & \text{if } n \text{ is even} \end{cases}$$

③ Find other percentiles - use definition $\textcircled{*}$

Examples:

$$\tilde{\pi}_{0.25}$$

- 25-th sample percentile
- first quartile
- bigger than 25% (one quarter) of all data

$$p = \frac{1}{4} \quad n = 11 \quad \frac{11}{4} \notin \mathbb{Z}, \quad \frac{11}{4} = 2 + 0.75$$

$$\tilde{\pi}_{0.25} = y_2 + \frac{3}{4} (y_3 - y_2) = \textcircled{62} = 59 + \frac{3}{4} (63 - 59)$$

$\tilde{\pi}_{0.75}$ - 75th sample percentile
- 3rd quartile

$$11 \cdot \frac{3}{4} = 8.25$$

$$\tilde{\pi}_{0.75} = y_8 + \frac{1}{4} (y_9 - y_8) = 83 + \frac{1}{4} \cdot 1 = \boxed{83.25}$$

New notations: • range = $y_n - y_1 = x_{\max} - x_{\min} = 41$

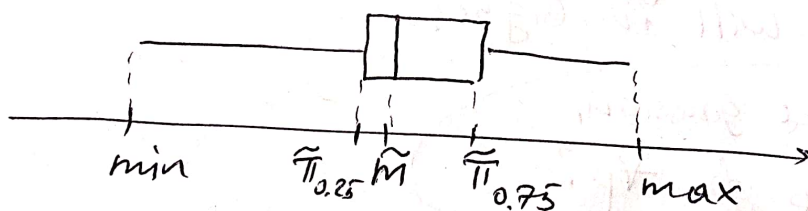
• IQR (interquartile range) $\stackrel{\text{def}}{=} \tilde{\pi}_{0.75} - \tilde{\pi}_{0.25} = 11.25$

• 5 number summary of the data

min; $\tilde{\pi}_{0.25}$; $\tilde{m}$; $\tilde{\pi}_{0.75}$; max

• outliers: $\{X_i: X_i < \tilde{\pi}_{0.25} - 3 \text{IQR} \text{ or } X_i > \tilde{\pi}_{0.75} + 3 \text{IQR}\}$

• Box plot



see pp 244-245
for more
versions of
box plots

Quantile-quantile plot (q-q plot)

— How do we know that data comes from like-gaussian distribution?

(it would be convenient:

- empirical rule needs ~~the~~ like-gaussian
- knowledge of theoretical cdf & pdf is super helpful - will see it later)

• histogram

• stem-and-leaf display

• q-q plot

α -q plot idea: compare gaussian percentiles (=quantiles) with sample percentiles

$\tilde{\pi}_p = (100p)$ percentile $= (p\text{-quantile}) = (100p)\%$ of all data is smaller than $\tilde{\pi}_p =$
 $= \underline{(1-p)\text{-fraction of data is bigger}}$

For gaussian distribution ~~each~~ "equivalent" point is $t_p: P(X > t_p) = 1-p$ $X \sim N(0,1)$

Such t_p is denoted as z_{1-p} and can be found for $\forall p$ in the gaussian table (see Table V6 in Appendix B)

Idea: if probability to be bigger than z_{1-p} is $1-p$ than roughly $(1-p)$ -fraction of independent samples will be bigger

if data is like-gaussian,
So, $\tilde{\pi}_p \approx z_{1-p}$ for $\forall p$?

Not really.

works only if mean of data is 0 and standard deviation is 1
(then $N(0,1)$ is correct scaling of gaussian)

However, for $N(\mu, \sigma^2)$

$\mu + \sigma z_{1-p} = q_{1-p}$
quantiles of a standard gaussian ($N(0,1)$) quantiles of a non-standard gaussian $N(\mu, \sigma^2)$

- Conclusion:
- if data is like-gaussian, $\tilde{\pi}_p$ and z_{1-p} must be in a linear relation
 - slope of this line is a sample standard deviation

Lifehack:

instead of plotting $(\tilde{\pi}_p, z_{1-p})$ for some random p 's,

- plot $(y_r, z_{1-\frac{r}{n+1}})$ for $r=1, \dots, n$ *

- see if they are almost on the line $\Rightarrow$ then like-gaussian

* since for
 $p = \frac{r}{n+1}; r = (n+1)p$
 $y_r = \tilde{\pi}_p$
 see Def *

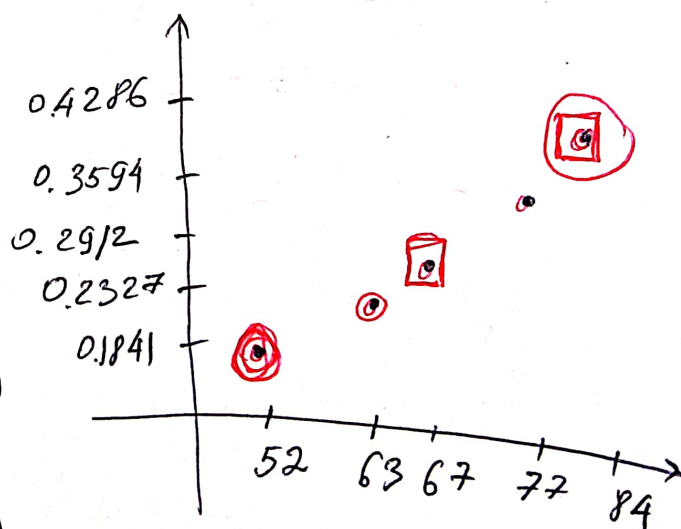
(Ex) $(52, z_{1-\frac{1}{11}}) = (52, z_{0.9})$

$(63, z_{1-\frac{2}{11}}) = (63, z_{0.73})$

$(67, z_{1-\frac{5}{11}}) = (67, z_{0.55})$

$(77, z_{1-\frac{7}{11}}) = (77, z_{0.36})$

$(84, z_{1-\frac{9}{11}}) = (84, z_{0.18})$



$z_\alpha: P(X > z_\alpha) = \alpha$